

DiffOBI: Diffusion-based Image Generation of Oracle Bone Inscription Style Characters

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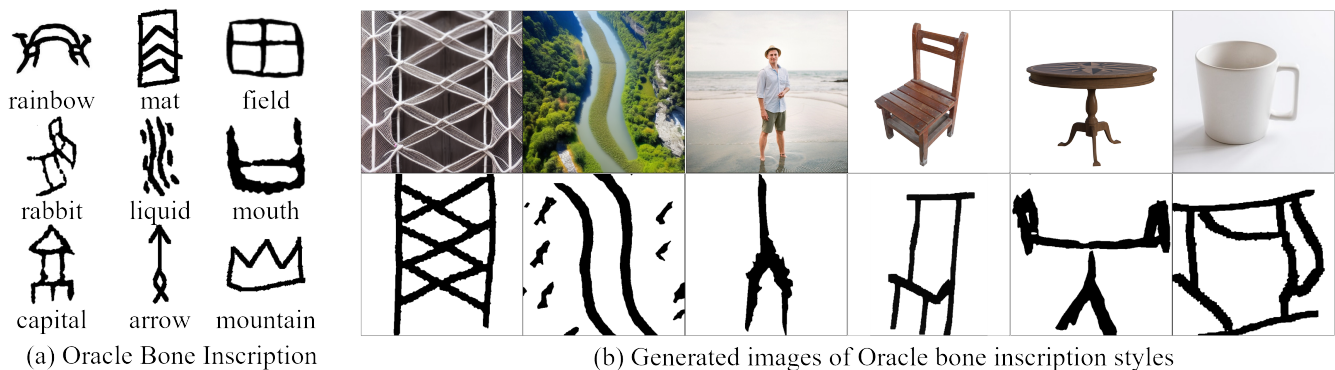


Figure 1: (a) Examples of oracle bone inscription from EVOBC dataset [Guan et al. 2024]. (b) Generated oracle bone inscription style images (second row) based on the input object images (first row) using DiffOBI.

Abstract

Oracle bone inscriptions are the most valuable asset of humankind with irreplaceable archaeological and visual values. However, it is still challenging to analyze and generate oracle bone inscription-style images for visual identification and art design. In this paper, we propose DiffOBI, a novel approach for generating images in the oracle bone inscription style using diffusion models. We first construct a dataset that aligns with oracle bone inscription, text prompts, and object images. This dataset serves as the foundation for fine-tuning ControlNet. By inputting images of various objects along with corresponding text prompts, the model generates the corresponding images in the style of oracle bone inscription. To further enhance the quality of the generated images, we integrate a refinement module to refine the initial results, ensuring the refined

results conform more closely to the original structure and norms of oracle bone inscription. This approach ensures that the generated images conform to the given input images and also preserves the unique pictographic features of oracle bone inscription.

CCS Concepts

• **Computing methodologies** → *Image processing*.

Keywords

Oracle bone inscription, image generation, diffusion model

ACM Reference Format:

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1 Introduction

Oracle bone inscriptions (OBIs), dating back over 3,500 years, are one of the oldest known writing systems in humankind history. Due to their scarcity and unique pictographic features, OBIs present challenges both in archeology and visual computation. Despite the discovery of thousands of oracle bone fragments, only 4,500 unique

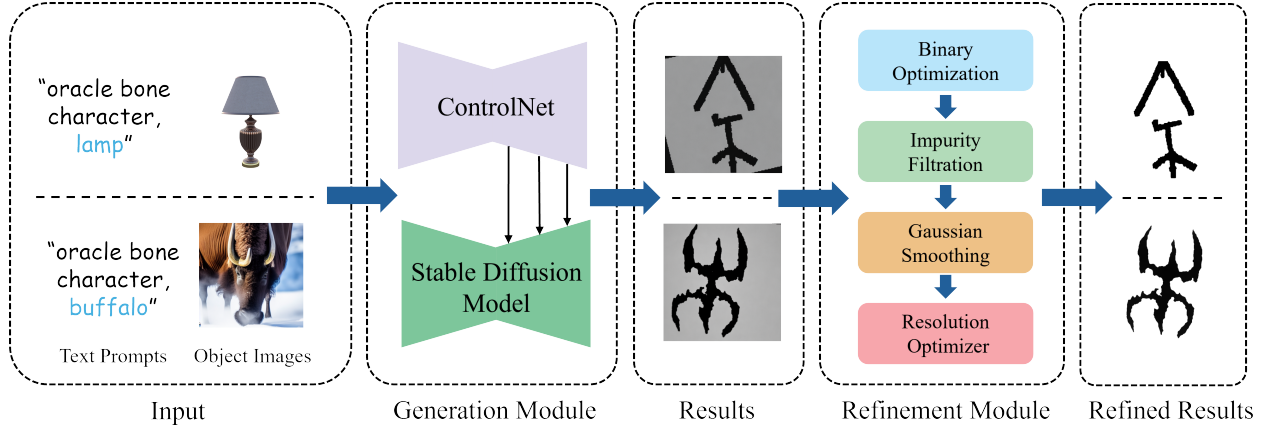


Figure 2: Framework of our method. Using text prompts and object images as inputs, we first generate the initial results via a generation module consisting of a stable diffusion model and ControlNet. In the refinement module, the initial results are refined for style, corners, edge, and resolution to obtain the refined results.

characters have been identified, with two-thirds still unidentified. Besides the visual identification work for OBIs, the OBI-style arts are a popular form of visual representation with the special abstractness and pictogram features¹. It is still a challenging issue to generate OBI-style images automatically from an image input.

OBIs lack a standardized writing system and are primarily preserved in image form. In addition, modern artifacts and contexts do not have corresponding historical inscriptions. It is difficult and impractical to directly generate OBIs-style images from input image features. To address these challenges, we adopt image generation model to generate OBI-style images from object images. Especially, we use diffusion models [Zhang and Agrawala 2023] which have shown significant potential in synthesizing high-quality images.

In this paper, we propose DiffOBI, a novel two-stage approach to generating oracle bone inscription-style images from given image inputs using diffusion models, ensuring the characters are stylistically accurate and semantically meaningful. We construct a dataset with images of the original oracle bone inscriptions, corresponding real-world images, and descriptive textual prompts. By inputting images of various objects along with corresponding text prompts, the model generates the corresponding images in the style of oracle bone inscriptions. To further enhance the quality of the generated images, we integrate a refinement module to improve the initial results, ensuring closer alignment with the original structure and norms of oracle bone inscriptions.

2 DiffOBI

Our proposed framework for OBI-style image generation consists of two stages (Fig. 2): generation stage and refinement stage.

In the generation stage, we utilize a fine-tuned image-to-OBI ControlNet [Zhang and Agrawala 2023] as an additional control and use a stable diffusion model to generate OBI-style images. These are integrated into a generation module, as shown in Fig. 2. The text prompts and object images are used as conditional inputs and are converted by the encoder into latent representations τ_θ .

Simultaneously, the object images are encoded into latent features c_f . During the diffusion process, random latent noise z_t is iteratively processed through the diffusion model $\phi_\theta(z_t, t, \tau_\theta)$ to obtain z_{t-1} , where t denotes each time step. By using c_f as an additional control condition, we can obtain the new representation as:

$$z_{t-1} = \phi_\theta(z_t, t, c_f, \tau_\theta) \quad (1)$$

By iterating ϕ_θ , we obtain $z_{t-1}, z_{t-2}, \dots, z_0$. The z_0 is decoded to generate the final OBI-style image as the initial result.

In the refinement stage, we enhance the initial images through Binary optimization, impurity filtration, edge refinement, and resolution enhancement. As shown in Fig. 2, these are combined into a refinement module. For binary optimization, pixel value detection and binarization are used to separate OBIs from the background:

$$P'(x, y) = \begin{cases} 0, & \text{if } P(x, y) < T \\ 255, & \text{if } P(x, y) \geq T \end{cases} \quad (2)$$

where $P(x, y)$ is the original pixel value, and $P'(x, y)$ is the binarized value, effectively distinguishing OBIs from the background. T denotes the threshold value. In our implementation, we used $T = 50$ after testing multiple threshold values ranging from 1 to 70, on a set of 100 generated images. This value consistently provided the best balance between separating the oracle bone inscriptions from the background and preserving their essential details.

For impurity filtration, we employ a mask-based filling method to remove dark corners that may appear in the initially generated results due to image rotation during data training. Specifically, we check the pixel values in the four corners of the binarized image, and if they are black, we remove the impurity interferences by replacing these black corner regions with white using a flood-filling method.

For edge refinement, we apply Gaussian smoothing as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (3)$$

where $G(x, y)$ represents the Gaussian function value at point (x, y) , σ is the standard deviation influencing the extent of smoothing, and x, y are the coordinates of a point relative to the kernel center.

¹<https://whatsonincapetown.com/exhibition-of-art-inspired-by-oracle-bones/>

For resolution enhancement, we apply bilateral filtering as follows:

$$I'(x) = \frac{1}{W_p} \sum_{x_i \in S} I(x_i) \cdot f_r(|I(x_i) - I(x)|) \cdot f_s(|x_i - x|) \quad (4)$$

where $I(x)$ is the original pixel intensity, $I'(x)$ is the filtered intensity, x_i denotes neighboring pixels, and S represents the spatial domain. The function f_r (range kernel) smooths intensity differences, f_s (spatial kernel) smooths spatial differences, and W_p is a normalization factor ensuring the sum of weights equals 1.

3 Experiment and Results

We constructed an OBIs image dataset. Specifically, we selected 36 sets of OBIs from the EVOBI dataset [Guan et al. 2024] with 5 images in each group, resulting in 180 original OBI images. Taking the meaning of these OBIs as prompts, we generated aligned images by the pre-trained ControlNet model [Zhang and Agrawala 2023], as shown in Fig. 3. This process resulted in a dataset consisting of 180 pairs of original OBI images, text prompts, and generated object images. 150 of these pairs are used for the training set, and 30 pairs are used for the test set. During the training phase, we fine-tuned the stable diffusion model (v2-1-512-ema-pruned) with ControlNet. Given the limited size of the dataset, we applied data augmentation techniques such as random flips, rotations, cropping, resizing, and adjustments to brightness, contrast, saturation, and tone. The model underwent 1,500 epochs of training with a batch size of 4 on NVIDIA A100 GPU.

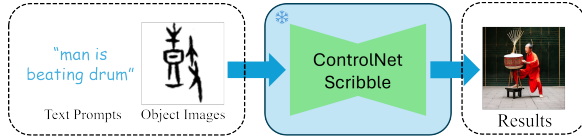


Figure 3: Original OBIs data and their meanings are used to generate aligned images via ControlNet.

3.1 Results

We evaluated our model from three aspects: fidelity in reconstructing existing items, capabilities to generate novel items not in the original dataset and produce varied styles for the same items.

For the reconstruction of existing characters, we compared our model's ability to replicate items from the dataset with IP-Adapter [Ye et al. 2023], pix2pix [Isola et al. 2018], and CycleGAN [Zhu et al. 2020]. The results, as shown in Fig. 4, demonstrate that our model effectively preserves the semantic and stylistic of OBIs.

For the generation of non-existent characters, we evaluated the model's capability to create images for items not present in the dataset. The images, illustrated in Fig. 5, effectively mirror the unique OBI style, demonstrating the model's ability to maintain stylistic integrity with novel objects.

For the evaluation of generation diversity, we examined the model's versatility in generating stylistic variations by using identical prompts for different styles of a single item (e.g., "oracle bone

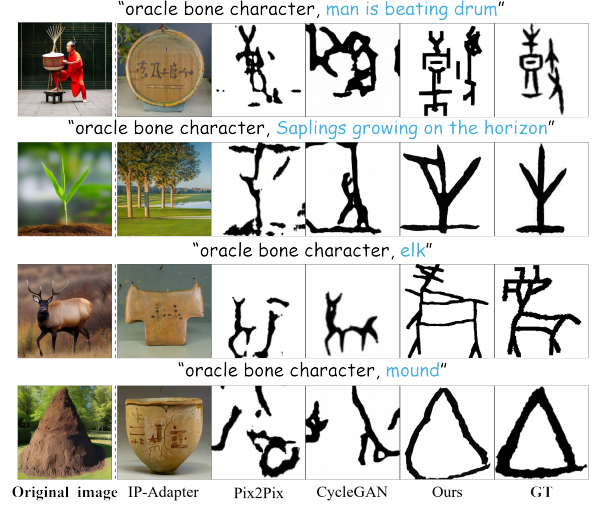


Figure 4: Results of generated images of objects (first row from original dataset, the others from the test dataset).

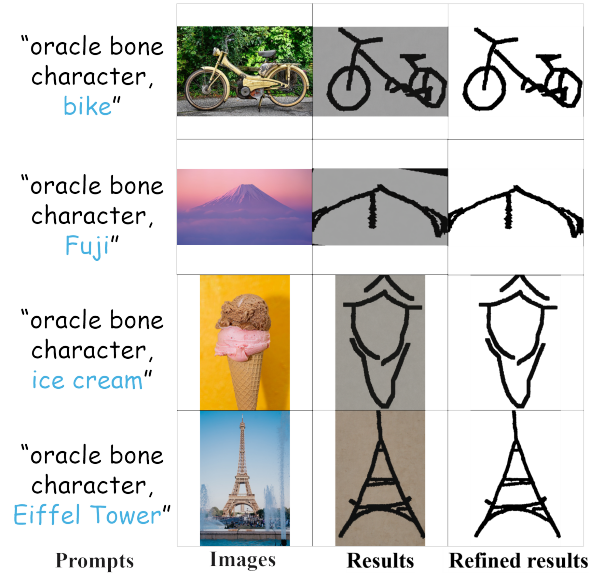


Figure 5: Results of generated images of objects not present in the original dataset.

character, lamp"). The outputs, shown in Fig. 6, successfully captured both the semantic essence and stylistic diversity of the inputs, while aligning with the traditional OBI aesthetic.

For quantitative evaluation, we evaluated the performance of IP-Adapter, pix2pix, CycleGAN, and our proposed model using three key metrics: Fréchet Inception Distance (FID), CLIP Image-Image similarity (CLIP-I), and Neural Image Assessment (NIMA). Table 2 shows our approach achieved the lowest FID score of 249.63, indicating the highest visual quality among all methods. In addition, our approach achieved the highest CLIP-I score of 0.81, suggesting

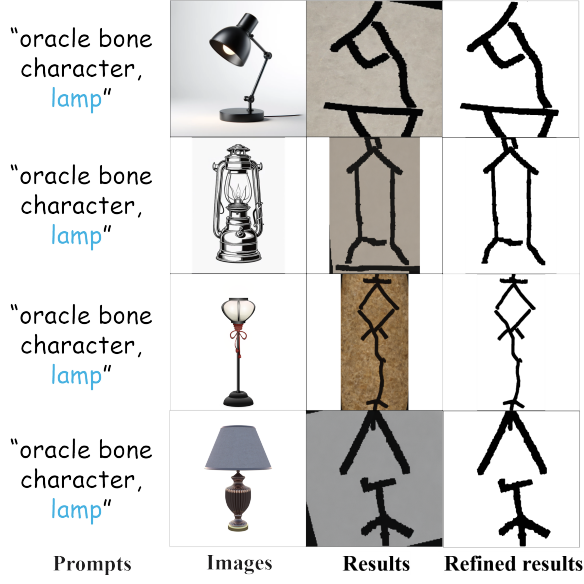


Figure 6: Examples of generated images showing stylistic variation within a single category.

Table 1: Quantitative comparison results.

	pix2pix	CycleGAN	IP-Adapter	Ours
FID (↓)	367.39	316.90	404.08	249.63
CLIP-I (↑)	0.79	0.76	0.56	0.81
NIMA (↑)	4.34	4.75	4.48	5.03

better alignment with the reference images. Our approach had NIMA score of 5.03, reflecting the best image aesthetic quality.

3.2 User Preference Study

We conducted a user preference study to evaluate our model against three generative models: IP-Adapter using diffusion models, pix2pix, and CycleGAN of GAN structures, each trained using the identical dataset. A visual comparison of the generated results from these models is shown in Fig. 7. The evaluation focused on two main criteria: fidelity to the OBIs style and similarity between the original and the generated OBI-style images. Initially, 100 participants familiarized themselves with the morphology of OBIs. They were then asked to select the four images that most closely represented this style from a set of 16 (four from each model). Our model was chosen in 44.00% of these selections, indicating a strong alignment with the oracle bone script style. Subsequently, participants compared the generated OBI-style images with their originals, rating the similarity on a scale from 1 (poor) to 5 (excellent). Our model achieved an average score of 3.54, outperforming the other models, as shown in Table 2.

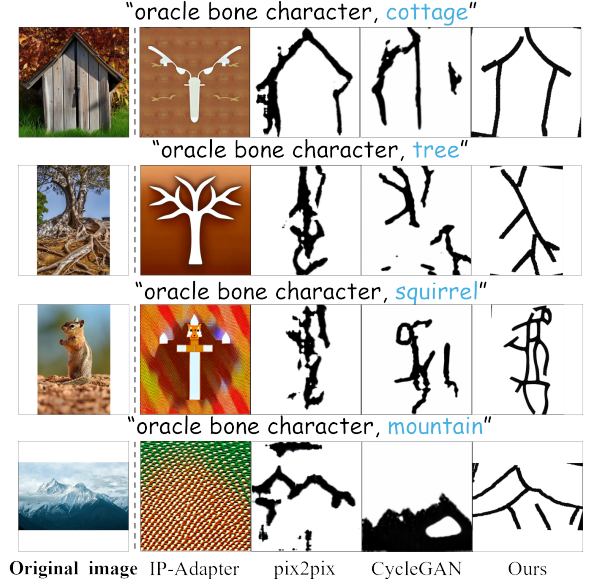


Figure 7: Visual Comparison of OBI-style images generated by IP-Adapter, pix2pix, CycleGAN, and ours. Note that pix2pix and CycleGAN did not require text prompts as input.

Table 2: User preference comparison results.

	pix2pix	CycleGAN	IP-Adapter	Ours
Selection rate/%	33.75	17.75	4.50	44.00
Average score	2.46	3.11	1.69	3.54

4 Conclusion

In this paper, we proposed DiffOBI, a two-stage approach using diffusion models to generate OBI-style images. Our experiment results show high visual quality and user preference, but there are some limitations. First, the model may struggle with multi-object images due to single-object-focused training data. Second, variations in brushstrokes and style may lead to discrepancies between the generated images and the ground truth. In future work, we plan to address these limitations by enhancing multi-object generation stability and exploring suitable evaluation metrics for diverse OBI styles. In addition, it is interesting to consider the evolution of OBI characters in the generation process.

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